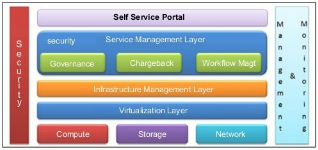

Figure 4: The building blocks of the private cloud

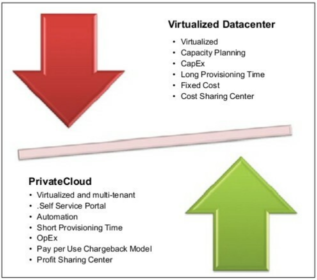

Figure 5: Virtualised environments vs private clouds

without the security concerns that come with sending data outside an organisation's boundary. This scenario appeals to organisations that desire more control over their infrastructure.

The building blocks of private clouds

The private cloud has come a long way, having gained significant traction in the market. Public cloud vendors who once criticised the private cloud as a 'false cloud' or as an oxymoron are now providing features for interoperability with private cloud platforms.

The decision to build your own cloud is a strategic one, which requires stakeholders to think about the many in-house resources required to sustain such infrastructure. It is the need of the hour for IT organisations to enable self-service, and become on-demand providers of infrastructure, platforms and applications for their internal business units. Speed and agility, not cost reduction, drive private cloud implementations.

Various cloud service providers offer the building blocks for a private cloud infrastructure–virtualisation (hypervisors such as ESXi, Xen, etc), self-service, metering or chargeback and automated workflow management.

Table 1: Open source private cloud product vendors

	OpenStack	Eucalyptus	CloudStack
Introduction	OpenStack offers open source software to build public and private clouds; started by Rackspace (Cloud Files platform) and NASA (Nebula) in 2010.	Eucalyptus is a short form of the Elastic Computing Architecture for Linking Your Programs to Useful Systems. It is an open source as well as a commercial private IaaS service provider.	CloudStack is open source cloud computing software for creating, managing and deploying Infrastructure as a Service (IaaS) clouds in service provider environments (the public cloud) or enterprise environments (the private cloud).
Components	Nova (compute), Swift (object storage), Glance (image service), Keystone (identity management), Horizon (GUI interface)	Cloud Controller (CLC): Manages the virtualisation resources and APIs; provides Web interface Walrus (S3 storage); Cluster Controller (CC): Controls execution of VMs and their networking; Storage Controller (SC): Provides block-level storage to VMs (EBS); Node Controller (NC): Controls VMs via hypervisors	Management server; Hypervisor nodes; Storage nodes; Layers: Zones, pods, clusters, hosts, primary storage, secondary storage
Hypervisor support	Xen, KVM, UML, LXC, VMware	Xen, KVM, VMware	Xen, KVM, VMware, Citrix XenServer
API availability	Yes	Yes	Yes
Self-service user portal	Yes	Yes (available with subscription to support)	Yes